

UNINVITED GUESTS FROM FACEBOOK

Putting doubts on Facebook's privacy settings once again is a birthday story of a 14 year old from Hertfordshire. She posted an invite, 21,000 said, "See you there..." Don't check "anyone can view" next time.
<http://bbc.in/97yjEa>

BEHIND THE SCENES OF NEWSY.COM

A 2-part story behind the formation of Newsy.com, a multi-source online video news site that monitors, analyzes and presents the world's news coverage. Also talks about Newsy apps for mobile internet devices.
<http://rww.to/bpAn0m>

THE NUMBER GAME

Gamers around the world have spent 2,000 years playing Bungie's latest installation Halo: Reach. All this even before Bungie celebrates the one week anniversary of the launch of the game. Wish people would spend that time trying to cure AIDS or cancer!
<http://bit.ly/9Plui0>

EXPLORE THE OCEANS WITH GOOGLE EARTH

Google Earth has come with its new app for the latest iOS. Built with the Apple retina display in mind, browsing Google Earth on the iPhone 4 is pure joy. Apart from surfing the planet's terrain now you can dive into the oceans, and experience the marine life.
<http://bit.ly/96k5eT>

market at whom Sandy Bridge is aimed, Intel will be packing the processors with an array of new media capabilities implemented with what the company calls a more easily programmable and flexible architecture.

These include dedicated hardware accelerators for tasks like high-quality video scaling, denoise filtering, deinterlacing, and film-mode detecting; the ability to hand more computes for processing high-definition workloads; high-quality enhancement techniques and filters for detail and adaptive contrast enhancement, and for total color control; native support of mainstream codecs like MPEG-2, VC1, and AVC, and technologies like stereoscopic 3D Blu-ray; and parallel engines that deliver high-throughput video rendering.

The EUs (shader units) are being optimized for the types of media expected to drive the next generation, and Sandy Bridge developers have worked to attain "parallel compute synergy" with the graphics processing hardware. This was done by more efficient media computation through low-power integer operations, more efficient vector/matrix ISA, and cooperation with the hardware accelerators. There's also language support for explicit parallel programming.

The result, says Intel, is that you can do all this faster while using lower power (and thus saving battery life on a laptop—an increasingly vital concern in a laptop-heavy computing world). Intel says processor power is cut in half for HD video playback, that most media applications consume about 30 percent less power at full rate than if they were using typical 3D, and that CPU-style power management achieves faster transition between power states.

There's one potential down side to this: If you add a discrete video card to computer,

you will lose all these added media capabilities. In some ways, that makes sense—after all, you generally lose access to integrated graphics capabilities when you're using a video card. But if the card you choose doesn't offer comparable media-management capabilities, you may want to seriously consider your own needs before you upgrade.

Turbo boost: boosted

One of the smarter innovations of recent Intel processors was Turbo Boost, which automatically increased the performance of certain cores by turning off others that weren't in use, assuming the electrical and thermal headroom were available. There's just one problem with this way of doing it: It assumes that the CPU reaches its maximum TDP immediately once turbo is enabled, while in actuality the CPU takes a little while to heat up.

So in Sandy Bridge, the PCU turbos active cores above the TDP for very short periods of time (up to about 25 seconds, according to Intel), stepping down gradually as the processor heats up until it finally reaches the real TDP. Because the PCU also tracks the available thermal budget when the system is idling, it can spend it once CPU demand increases. (Like current Turbo Boosting, however, the PCU will not allow the chip to exceed any hard reliability limits.)

Conclusion: a bridge to the future?

It's impossible to say right now whether Sandy Bridge will live up to Intel's extravagant claims for it—we still have a few months left until it's released, and a lot can happen in that time.

But based on what we've seen and heard so far, we're intrigued. It looks like Sandy Bridge doesn't just build on Nehalem's innovations,

but it also expands into directions we hadn't previously considered. Intel's techniques for dealing with the on-board graphics (particularly media processing) and the revamped Turbo Boost strike us as particularly exciting changes, and we can't wait to see how—if?—they pan out in the real world.

We want to reserve most of our judgments until Sandy Bridge is released, but we have a few qualms. First, we find it a bit odd that the CPUs will support only DX10.1 graphics and not DX11 out of the gate. Considering that DX11 will have been out for over a year when Sandy Bridge is released, this seems like a significant oversight—and it's one Intel can count on AMD hammering at every opportunity, especially given that its own Fusion APUs will be out around the same time.

It's also a little disappointing that, in order to take advantage of all of Sandy Bridge's enhancements, owners of computers with the previous mainstream next-generation socket, the LGA1156, will have to upgrade yet again, to the LGA1155. So much of what Sandy Bridge brings to the table has the potential of really benefiting mainstream users—it's too bad owners of current (and, let's face it, recently purchased or build systems) won't just be able to pop in the new chips.

Of course, these could all be moot points if Sandy Bridge proves to be so spectacular as to be worth the upgrade. And for that, again, we have to wait—as do enthusiasts, who won't be reaping the benefits of early adoption this time around. Intel is taking a lot of surprising chances, and making a lot of daring choices, with Sandy Bridge, and seeing whether they pay off—and how they compare to AMD's own ideas about uniting formerly disparate technologies on the same processor die—is already poised to be one of the most exciting tech events of early 2011.